

A rare-event obstruction to the uniform exponential GEPP tail in IE-04

An explicit robust-growth construction and a Gaussian small-box bound

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1 Statement and conclusion

IE-04 asks for universal $c_1, c_2 > 0$ such that, for every admissible dimension, deterministic center, noise level, and $x \geq 1$,

$$\Pr\{\rho_{\text{PP}}(\bar{A} + \sigma G) > x(n/\sigma)^{c_1}\} \leq 2^{-c_2 x}, \quad (1)$$

where $\|\bar{A}\|_2 \leq 1$, $0 < \sigma \leq 1$, and G has independent real standard normal entries [1]. The growth factor is the largest absolute entry among all active Schur complements, divided by the largest absolute input entry. Elimination is in exact arithmetic.

Theorem. For every $n \geq 2$ and either $C = 0$ or $C = I_n$, a standard Gaussian matrix G satisfies

$$\Pr\left\{\rho_{\text{PP}}(C + G) > \frac{1}{2} \left(\frac{3}{2}\right)^{n-1}\right\} \geq 2^{-n^2(n^2+n+5)} \geq 2^{-3n^4}. \quad (2)$$

Consequently, (1) is false for every choice of positive constants c_1, c_2 . The case $\bar{A} = I_n$, $\sigma = 1$ already contradicts it. This deterministic center is nonsingular and has spectral norm one.

The point is the requirement of an exponential tail in **every** $x \geq 1$. An exponentially large growth factor has at least an inverse-exponential-in-a-polynomial-of- n probability; the proposed right-hand side at the corresponding x is much smaller. This does not deny a polynomial growth bound with high probability, or settle a different tail estimate restricted to a smaller range of x .

2 A strictly pivoted high-growth matrix

Define $W_n = (w_{ij})$ by

$$w_{ij} = \begin{cases} 1, & j = n, \\ 1, & j < n \text{ and } i = j, \\ -1/2, & j < n \text{ and } i > j, \\ 0, & j < n \text{ and } i < j. \end{cases} \quad (3)$$

For example, its order-four instance is

$$W_4 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1/2 & 1 & 0 & 1 \\ -1/2 & -1/2 & 1 & 1 \\ -1/2 & -1/2 & -1/2 & 1 \end{pmatrix}.$$

At elimination step $k < n$, the pivot is 1 and every competitor in the pivot column is $-1/2$. Thus partial pivoting uniquely chooses the current row. The other nonfinal columns keep the same triangular pattern. In the final column, all remaining entries have a common value c_k , and a step changes this to $c_k - (-1/2)c_k = (3/2)c_k$. Hence

$$c_k = \left(\frac{3}{2}\right)^{k-1}, \quad \rho_{\text{PP}}(W_n) = c_n. \quad (4)$$

In particular, W_n is nonsingular. All entries of every exact active matrix have absolute value at most 2^n .

3 Quantitative robustness on a full entrywise box

Set

$$B_n = 2^{n+2}, \quad \delta_n = 2^{-(n^2+n+1)}. \quad (5)$$

These constants satisfy

$$B_n^{n-1} \delta_n = \frac{1}{8}. \quad (6)$$

Lemma. Every real matrix A with $\|A - W_n\|_{\max} \leq \delta_n$ is nonsingular, has a unique no-swap partial-pivoting path, and satisfies

$$\rho_{\text{PP}}(A) > \frac{c_n}{2}. \quad (7)$$

Proof. Denote the active Schur complements of W_n and A by S_k and T_k . We prove by induction that the same no-swap path is selected and

$$e_k := \|T_k - S_k\|_{\max} \leq B_n^{k-1} \delta_n \leq \frac{1}{8}. \quad (8)$$

The initial claim follows from the box condition. Suppose it holds at step $k < n$; retain the original global indices on the active entries. The perturbed pivot has value at least $1 - e_k \geq 7/8$, while every competing entry has absolute value at most $1/2 + e_k \leq 5/8$. The pivot is therefore strictly largest and positive.

Write $b = -1/2$ for the unperturbed elimination multiplier in a remaining row, and $q = (T_k)_{ik}/(T_k)_{kk}$ for its perturbed value. For $e = e_k \leq 1/8$,

$$|q| \leq \frac{1/2 + e}{1 - e} \leq \frac{5}{7} < 1, \quad |q - b| \leq \frac{(3/2)e}{1 - e} \leq 2e. \quad (9)$$

For each remaining (i, j) , subtracting the two exact update formulas gives

$$(T_{k+1} - S_{k+1})_{ij} = (T_k - S_k)_{ij} - q(T_k - S_k)_{kj} - (q - b)(S_k)_{kj}. \quad (10)$$

Using $|(S_k)_{kj}| \leq 2^n$ and (9),

$$e_{k+1} \leq (2 + 2^{n+1})e_k \leq B_n e_k. \quad (11)$$

This proves the induction, including the strict pivot selection at every nonfinal step. The final scalar Schur complement is at least $c_n - 1/8 > 0$, so all pivots are nonzero and A is nonsingular. Also $\|A\|_{\max} \leq 1 + \delta_n \leq 9/8$. Consequently,

$$\rho_{\text{PP}}(A) \geq \frac{c_n - 1/8}{1 + \delta_n} \geq \frac{8c_n - 1}{9} \geq \frac{7c_n}{9} > \frac{c_n}{2}, \quad (12)$$

where $c_n \geq 1$ was used. This proves the lemma. Notice that the statement concerns the entire box, not sampled perturbations, and uses no unspecified continuity radius.

4 Gaussian probability of the box

Fix $C \in \{0, I_n\}$ and set $A = C + G$. Define the event

$$E_n = \{ |G_{ij} - (w_{ij} - C_{ij})| \leq \delta_n \text{ for every } i, j \}. \quad (13)$$

On E_n , the matrix $A = C + G$ belongs to the box around W_n in the lemma. For the smoothed counterexample we use $C = I_n$, so $\bar{A} = I_n$ and $\sigma = 1$. This obeys $\|\bar{A}\|_2 = 1$ and even a requirement that the deterministic center be nonsingular. The matrix W_n is only the center of the high-growth event in input space; there is no requirement that $\|W_n\|_2 \leq 1$. The case $C = 0$ gives an additional pure-Gaussian statement but is not needed for the counterexample.

Each interval in (13) lies within $[-2, 2]$, because $|w_{ij} - C_{ij}| \leq 1$ for both choices of C , and $\delta_n \leq 1/8$. The standard normal density there is bounded below by

$$\frac{e^{-2}}{\sqrt{2\pi}} > \frac{1}{32}. \quad (14)$$

For instance, $e < 3$ and $\pi < 4$ imply that the left side exceeds $1/27$. Each interval has length $2\delta_n$, and the entries are independent. It follows that

$$\Pr(E_n) \geq \left(\frac{\delta_n}{16}\right)^{n^2} = 2^{-n^2(n^2+n+5)}. \quad (15)$$

The lemma gives $E_n \subseteq \{\rho_{\text{PP}}(C + G) > c_n/2\}$. Finally, $n^2(n^2 + n + 5) \leq 3n^4$ for $n \geq 2$: subtracting gives $n^2(2n^2 - n - 5) > 0$. This proves (2).

5 Contradiction for arbitrary proposed constants

Let $c_1, c_2 > 0$ be arbitrary. Put

$$x_n = \frac{(3/2)^{n-1}}{2n^{c_1}}, \quad K_n = n^2(n^2 + n + 5). \quad (16)$$

Exponential growth dominates every fixed power of n , so $x_n \geq 1$ and $c_2 x_n > K_n$ for all sufficiently large n . More explicitly,

$$\log \frac{x_n}{K_n} = (n-1) \log(3/2) - (c_1 + 4) \log n - \log 2 - \log(1 + 1/n + 5/n^2) \longrightarrow +\infty. \quad (17)$$

For such n , (2) with $C = I_n$ yields

$$\Pr\{\rho_{\text{PP}}(I_n + G) > x_n n^{c_1}\} \geq 2^{-K_n} > 2^{-c_2 x_n}, \quad (18)$$

contradicting (1) in an admissible instance. Thus no pair of universal positive constants can satisfy the displayed IE-04 inequality.

6 Verification, scope, and provenance

This is an analytic proof for every dimension and every proposed pair of constants. The accompanying `verify_bounds.py` supplies additional exact checks: the default run certifies the **whole** entrywise box for each $n = 2, \dots, 16$ by outward-rounded dyadic interval arithmetic, verifies the scalar bounds for $n = 2, \dots, 256$, and gives illustrative exact contradiction dimensions for several selected constants. These finite checks do not replace the universal argument above. They require only Python’s standard library.

`python verify_bounds.py`

`certificate.json` and `verification.txt` record the successful run. The earlier interrupted work contained the high-growth/small-box proof outline; this explicit version, including the constants and all inequalities, was completed and checked during recovery. The argument is submitted as an AI-assisted proof for review. No external human peer review, formal proof-assistant certification, or novelty/priority determination is claimed.

The conclusion concerns precisely (1), with its unrestricted $x \geq 1$. It does not produce an alternative optimal smoothed tail, resolve growth expectations, or identify a sharp high-probability polynomial exponent. The historical motivation appears in [2], Conjecture 16; the canonical repository formulation is the target certified here.

7 References

- [1] OpenProblemsInNLA, IE-04, canonical statement, displayed smoothed probability inequality and growth convention; accessed during the recovery session. Repository entry.
- [2] D. A. Spielman and S.-H. Teng, *Smoothed Analysis of Algorithms and Heuristics: Progress and Open Questions*, in *Foundations of Computational Mathematics, Santander 2005* (2006), 274–342. Section P6, Conjecture 16, page 52 of the author PDF. DOI: 10.1017/CBO9780511721571.010.